# Runbo Li

## *Curriculum Vitae*

✉ *runbo.li26@imperial.ac.uk*
✉ *runbo.li.carey@gmail.com*
*https://runbolicarey.com*
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## Education

2026 – **B.Sc.** Department of Mathematics, Imperial College London, South Kensington Campus, London SW7 2AZ, UK

2023 – 2026 International Curriculum Center (ICC), The High School Affiliated to Renmin University of China, Beijing, China

## Publications

6 **Runbo Li**. A note on variants of Buchstab's identity. *Bulletin of the Belgian Mathematical Society – Simon Stevin*. to appear.

5 **Runbo Li**. On a conjecture involving twin practical numbers. *Notes on Number Theory and Discrete Mathematics*. 32(3): 544–555, 2026. *https://doi.org/10.7546/nntdm.2026.32.3.544-555*.

4 **Runbo Li**. On the largest prime factor of integers in short intervals. *Cambridge Journal for Junior Scientists*. 3(1): 1–16, 2026. *https://doi.org/10.4310/CJJS.260205233711*.

3 **Runbo Li**. A remark on the distribution of $\sqrt{p}$ modulo one involving primes of special type. *Hiroshima Mathematical Journal*. 56(1): 57–60, 2026. *https://doi.org/10.32917/h2024011*.

2 **Runbo Li**. On Chen's theorem, Goldbach's conjecture and almost prime twins II. *Mathematical Reports*. 28(1–2): 39–61, 2026. *https://doi.org/10.59277/mrar.2026.28.78.1.2.39*.

1 **Runbo Li**. A remark on the distribution of $\sqrt{p}$ modulo one involving primes of special type II. *European Journal of Mathematics*. 11:85, 2025. *https://doi.org/10.1007/s40879-025-00877-y*.

## Preprints

39 **Runbo Li**. Waring–Goldbach Problem: One square, four fifth powers and twenty-two seventh powers. *Hexagon*. *Hexagon:2610.00423v1*.

38 **Runbo Li**. Waring–Goldbach Problem: One square, four fifth powers and fifteen sixth powers. *Hexagon*. *Hexagon:2610.00422v1*.

37 **Runbo Li**. Note on variable role-reversals. *Hexagon*. *Hexagon:2610.00380v1*.

36 **Runbo Li**. Large value estimates for Dirichlet polynomials and trilinear sums over short intervals. *arXiv*. *arXiv:2610.18923v1*.

35 **Runbo Li**. On the largest prime factor of integers in short intervals IV. *Cambridge Open Engage*. *10.33774/coe-2026-24kjf*.

34 **Runbo Li**. The number of Piatetski-Shapiro primes in short intervals. *Cambridge Open Engage*. *10.33774/coe-2026-qpr94*.

33 **Runbo Li**. Primes in arithmetic progressions to large moduli and refinements of Harman's sieve. *arXiv*. *arXiv:2602.20917v6*.

32 **Runbo Li**. On the largest prime factor of integers in short intervals III. *arXiv*. *arXiv:2601.00910v1*.

31 **Runbo Li**. Sums of two squares of primes in short intervals. *Cambridge Open Engage*. *10.33774/coe-2025-15m3h*.

30 **Runbo Li**. Bombieri–Vinogradov Theorem in shorter intervals. *Preprints*. *preprints202511.0089.v1*.

29 **Runbo Li**. Primes in almost all short intervals II. *Cambridge Open Engage*. *10.33774/coe-2025-jrnjl*.

28 **Runbo Li**. On Chen's theorem, Goldbach's conjecture and almost prime twins III. *Cambridge Open Engage*. *10.33774/coe-2025-5kl50*.

27 **Runbo Li**. An average Brun–Titchmarsh theorem and shifted primes with a large prime factor. *arXiv*. *arXiv:2508.18285v3*.

26 **Runbo Li**. On the weighted AM-GM inequality and refined inequalities between arithmetic functions. *Preprints*. *preprints202511.0246.v1*.

25 **Runbo Li**. On the exceptional set in the $abc$ conjecture. *arXiv*. *arXiv:2507.02885v1*.

24 **Runbo Li**. On the largest prime factor of integers in short intervals II. *Cambridge Open Engage*. *10.33774/coe-2025-xnbjq-v2*.

23 **Runbo Li**. On Chen's theorem, Goldbach's conjecture and almost prime twins II. *arXiv*. *arXiv:2405.05727v4*.

22 **Runbo Li**. On the largest prime factor of quadratic polynomials. *arXiv*. *arXiv:2406.07575v2*.

21 **Runbo Li**. A remark on the distribution of $\sqrt{p}$ modulo one involving primes of special type II. *arXiv*. *arXiv:2401.01351v3*.

20 **Runbo Li**. Largest square divisors of shifted primes. *arXiv*. *arXiv:2505.23779v2*.

19 **Runbo Li**. On the generalized Dirichlet divisor problem. *Preprints*. *preprints202505.1432.v3*.

18 **Runbo Li**. On almost primes in Piatetski-Shapiro sequences. *arXiv*. *arXiv:2505.09634v1*.

17 **Runbo Li**. Primes in arithmetic progressions to smooth moduli: A minorant version. *arXiv*. *arXiv:2505.09629v3*.

16 **Runbo Li**. On prime-producing sieves and distribution of $\alpha p - \beta \mod 1$. *arXiv*. *arXiv:2504.13195v3*.

15 **Runbo Li**. A note on variants of Buchstab's identity. *arXiv*. *arXiv:2504.07974v1*.

14 **Runbo Li**. The number of primes in short intervals and numerical calculations for Harman's sieve. *arXiv*. *arXiv:2308.04458v8*.

13 **Runbo Li**. On the Piatetski-Shapiro prime number theorem II. *Preprints*. *preprints202504.1165.v2*.

12 **Runbo Li**. On the Piatetski-Shapiro prime number theorem. *Preprints*. *preprints202504.1190.v1*.

11 **Runbo Li**. On the largest prime factor of integers in short intervals. *Preprints*. *preprints202504.1212.v2*.

10 **Runbo Li**. Primes in almost all short intervals. *arXiv*. *arXiv:2407.05651v6*.

9 **Runbo Li**. On the primitive divisors of quadratic polynomials. *arXiv*. *arXiv:2406.07575v1*.

8 **Runbo Li**. On Chen's theorem, Goldbach's conjecture and almost prime twins. *arXiv*. *arXiv:2405.05727v3*.

7 **Runbo Li**. On a conjecture involving twin practical numbers. *Preprints*. *preprints202504.1211.v2*.

6 **Runbo Li**. A remark on large even integer of the form $p + P_3$. *arXiv*. *arXiv:2403.09691v1*.

5 **Runbo Li**. Hybrid estimation of exponential sums, exceptional characters and primes in short intervals. *arXiv*. *arXiv:2401.11139v3*.

4 **Runbo Li**. Remarks on additive representations of natural numbers. *arXiv*. *arXiv:2309.03218v7*.

3 **Runbo Li**. A remark on the distribution of $\sqrt{p}$ modulo one involving primes of special type. *arXiv*. *arXiv:2401.01351v1*.

2 **Runbo Li**. On the upper and lower bound orders of almost prime triples. *arXiv*. *arXiv:2401.01348v1*.

1 **Runbo Li**. On some problems of primes with the floor function. *arXiv*. *arXiv:2308.16301v1*.

## Manuscripts

1 **Runbo Li**. Primes in almost all short intervals III.

## Presentations

6 Primes in arithmetic progressions to large moduli. *The 4th Workshop on Cute Math*. Online. 26 July 2026.

5 Primes in short intervals. *Nankai University Additive Number Theory Seminar*. Online. 4 December 2025.

4 On Bertrand's Postulate. *Tianjin Nankai High School: Student Research Experience Sharing and Shiing-Shen Chern Class Research Group Launch Meeting*. Online. 21 November 2025.

3 Primes in short intervals. *The 3rd Workshop on Cute Math*. Online. 21 July 2025.

2 On the Goldbach's conjecture. *2025 'Haidian $\pi$ Mathematics Festival' Closing Ceremony Research Presentation*. The High School Affiliated to Beijing Institute of Technology. 4 July 2025.

1 On Chen's theorem, Goldbach's conjecture and applications of sieve methods. *S.-T. Yau High School Science Award (Mathematics) Global Final Research Report*. Tsinghua University. 7 December 2024.

## Attended Seminars and Coferences (Selected)

2026 2026 Chen Jing-Run Prize Conference (Beijing, China)

2026 Forum for young scholar in number theory and related fields (Xiamen, China)

2024 2024 Number Theory Workshop at Peking (The 9th National Conference on Number Theory) (Beijing, China)

2024 International Number Theory Conference in Commemoration of Chengdong Pan (Jinan, China)